

Chapter 9: Applications of Fixed Point Theorems

Applications to Integral and Volterra Equations, Surjectivity Theorems

Based on Pathak: Introduction to Nonlinear Analysis and Fixed Point Theory

Pages 738–757

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Section 9.5: Nonlinear Integral Equations

Problem Framework

We consider the integral equation of the form

$$x(t) = u(t, x(t)) + \int_0^t p(t, s, x(s)) ds, \quad t \in [0, a]$$

where

- $u : [0, a] \times \mathbb{R} \rightarrow \mathbb{R}$ is continuous
- $p : [0, a] \times [0, a] \times \mathbb{R} \rightarrow \mathbb{R}$ is continuous
- $g : [0, a] \times [0, a] \rightarrow \mathbb{R}^+$ is a continuous bound

This is a classical nonlinear Volterra integral equation that arises in many applied problems.

Fixed Point Formulation

Mapping on $C[0, a]$

Define the operator $T : C[0, a] \rightarrow C[0, a]$ by

$$(Tx)(t) = u(t, x(t)) + \int_0^t p(t, s, x(s)) ds$$

Finding a solution to the integral equation is equivalent to finding a fixed point of T .

Key Step: Show that T satisfies the **Darbo condition**:

$$\mu(TX) \leq q\mu(X) \quad \text{for some } q < 1$$

where μ denotes the Kuratowski measure of noncompactness.

Assumptions for Existence

Conditions A1–A4

A1: There exist constants $k, k', a\beta$ and a nondecreasing function $\phi : [0, \infty) \rightarrow [0, \infty)$ with $\phi(0) = 0$ such that

$$|u(t, x) - u(t, y)| \leq k|x - y|, \quad |p(t, s, x) - p(t, s, y)| \leq a\phi(|x - y|)$$

for all $t, s \in [0, a]$ and $x, y \in \mathbb{R}$.

A2: $|u(t, 0)| \leq m$ and $|p(t, s, 0)| \leq \psi(t, s)$ for some $m \geq 0$ and measurable function ψ .

A3: $\gamma = k + ka\beta < 1$.

A4: $\eta = ka\ell + m$ where $\ell = \max_{t,s} |g(t, s)|$.

Contraction Property of T

Darbo Condition Analysis

From assumptions A1–A4, we can show that the operator T maps the ball

$$B_r = \{x \in C[0, a] : \|x\| \leq r\} \quad \text{into itself}$$

where $r_1 \leq r \leq r_2$ with

$$r_1 = \frac{(1 - 2y\eta) - \sqrt{1 - 4y\eta}}{2y^2}, \quad r_2 = \frac{(1 - 2y\eta) + \sqrt{1 - 4y\eta}}{2y^2}$$

Furthermore, the operator T satisfies the Darbo condition with respect to the measure ω_0 :

$$\omega_0(TX) \leq 2k\omega_0(X), \quad \omega_0(GX) \leq 2k\omega_0(X)$$

Hence T is a contraction with respect to ω_0 and has at least one fixed point.

Special Cases of Integral Equations I

The equation $x(t) = u(t, x(t)) + \int_0^t p(t, s, x(s))ds$ encompasses many classical forms:

Case 1: If $g(t, y, x) = 1$, then

$$x(t) = u(t, x(t)) + \int_0^t p(t, s, x(s))ds \quad [\text{Ref. 380}]$$

Case 2: For $u(t, x) = 0$, we get the nonlinear functional-integral equation

$$x(t) = f\left(t, \int_0^t p(t, s, x(s))ds\right) \cdot g\left(t, \int_0^a q(t, s, x(s))ds\right) \quad [\text{Ref. 38, 379}]$$

Case 3: The functional-integral equation

$$x(t) = u(t, x(t)) + \int_0^t p(t, s, x(s))ds \quad [\text{Ref. 37}]$$

Special Cases of Integral Equations II

Case 4: With $u(t, x) = 0, g(t, y, x) = 1, f(t, y, x) = u(t, x)y$:

$$x(t) = u(t, x(t)) \int_0^t p(t, s, x(s)) ds \quad [\text{Ref. 381}]$$

Classical Integral Equations I

Volterra, Urysohn, and Quadratic

Case 5: Classical Volterra integral equation (with $u = 0, g = 1, f = 1$):

$$x(t) = a(t) + \int_0^t p(t, s, x(s)) ds \quad [\text{Standard form}]$$

Case 6: Urysohn integral equation (with $u = 0, f = 1, g = b(t) + y$):

$$x(t) = b(t) + \int_0^a q(t, s, x(s)) ds$$

Case 7: Composite functional-integral equation (with $u = 0, f = a(t) + y, g = y$):

$$x(t) = a(t) \int_0^a q(t, s, x(s)) ds + \left(\int_0^t p(t, s, x(s)) ds \right) \left(\int_0^a q(t, s, x(s)) ds \right)$$

Classical Integral Equations II

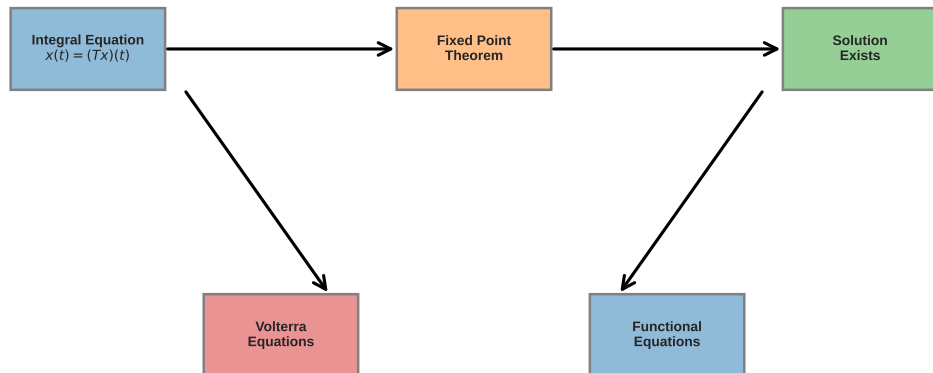
Volterra, Urysohn, and Quadratic

Case 8: Chandrasekhar quadratic integral equation (if $\phi(s)$ is even polynomial):

$$x(t) = 1 + x(t) \int_0^a \frac{t}{t+s} \phi(s) x(s) ds \quad [\text{Chandrasekhar type}]$$

Application Framework

Visual Summary



Section 9.6: Volterra Integrodifferential Equations

Problem Statement

Consider the system of Volterra integrodifferential equations:

$$\dot{u}(t) = H(T, t_0, f, g, h, K_1, K_2), \quad t \geq t_0 \quad (1)$$

$$\dot{\bar{u}}(t) = H(T, t_0, \bar{f}, \bar{g}, \bar{h}, \bar{K}_1, \bar{K}_2), \quad t \geq t_0 \quad (2)$$

where H is a nonlinear functional involving:

- The time variable t and initial time t_0
- Vector fields f, g, h and their variants
- Kernels $K_1(t), K_2(t)$ (with exponential bounds)

Goal: Prove existence and uniqueness of mild solutions in appropriate function spaces.

Function Space Setup

The Space $C([t_0, t_1], B)$

Define the normed space

$$C([t_0, t_1], B) = \left\{ u \in C([t_0, t_1], B) : \|u\|_C = \max_{t \in [t_0, t_1]} \|u(t)\| \right\}$$

Key properties:

- $C([t_0, t_1], B)$ is a Banach space
- Solutions $u(t)$ are continuous functions with $u(t_0) = u_0 \in B$
- The Lipschitz condition ensures contraction on this space

Theorem 9.34 (Pathak): Under suitable Lipschitz and boundedness conditions, the initial value problems (9.149) and (9.150) have a unique common mild solution $u(t) \in C([t_0, t_1], B)$.

Key Lemma for Volterra Equations

Lemma 9.5 (Pachpatte)

Let $x(t), a(t), b(t), c(t)$ be nonnegative continuous functions on $\mathbb{R}^+$ satisfying

$$x(t) \leq x_0 + \int_0^t a(s)x(s)ds + \int_0^t a(s) \left(\int_0^s b(r)x(r)dr \right) ds + \int_0^t a(s) \left(\int_0^s b(r) \left(\int_0^r c(z)x(z)dz \right) dr \right) ds$$

Then

$$x(t) \leq x_0 \left(1 + \int_0^t a(s) \exp \left(\int_0^s a(r)dr \right) \left[1 + \int_0^s b(r) \exp \left(\int_0^r (b(z) + c(z))dz \right) dr \right] ds \right)$$

This bounds the growth of solutions and is crucial for proving uniqueness.

Existence and Uniqueness Theorem I

Theorem 9.34

Theorem 9.34 (Pathak): Suppose conditions A_1 – A_4 are satisfied. Then, for $u_0 \in B$, the initial value problems (9.149) and (9.150) have a unique common mild solution $u(t) \in C([t_0, t_1], B)$ for $t \geq t_0$, where t_1 is arbitrarily fixed with $t_1 > t_0$.

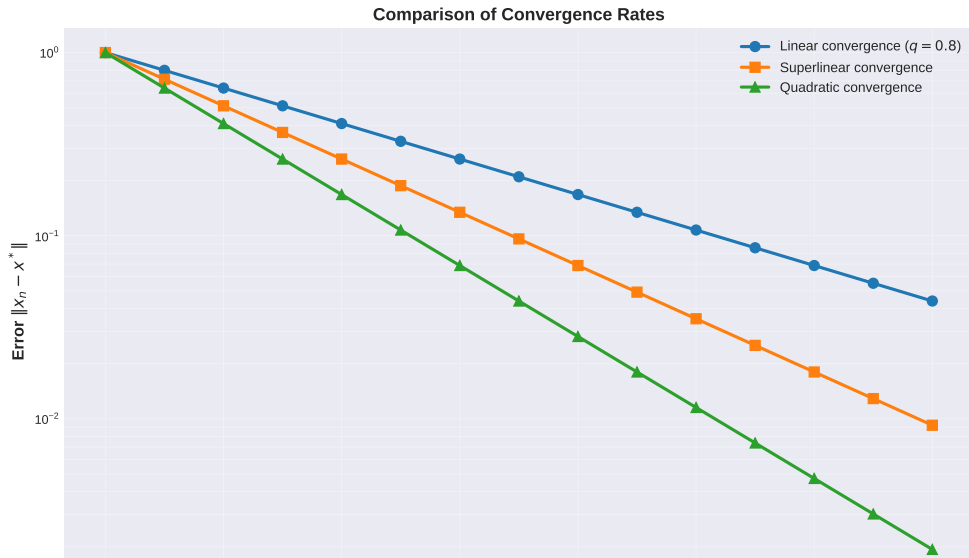
Moreover, the map $u_0 \mapsto u$ from B into $C([t_0, t_1], B)$ is Lipschitz continuous.

Proof Sketch:

- 1 Define operators $F, \bar{F} : C \rightarrow C$ by the right-hand sides of (9.152) and (9.153)
- 2 Show that F and $\bar{F}$ map C into itself using $\|Fx\| \leq M$ for all $x \in C$
- 3 A common fixed point of F and $\bar{F}$ gives a common mild solution
- 4 Use boundedness of the Lipschitz constant to ensure contraction
- 5 Apply Banach's fixed point theorem

Convergence Behavior

Comparison of Rates



Section 9.7: Surjectivity Theorems

Generalized Directional Contractors

Let X and Y be Banach spaces. Let $P : \mathcal{D}(P) \subset X \rightarrow Y$ have closed graph.

Definition: We define the class (S) of generalized directional contractors. Let $\Gamma(x)(P)$ be the set of generalized directional contractors for P at $x \in \mathcal{D}(P)$ called class (S) if:

- There exists $q = q(P) < 1$, a constant $B > 0$
- There is a nondecreasing function $c : [0, \infty) \rightarrow [0, \infty)$ with $c(0) = 0$
- For each $y \in \Gamma(x)(P)$, there exist $\varepsilon = \varepsilon(x, y) \leq 1$ and $\bar{x} \in \mathcal{D}(P)$ such that:

$$(S_1) \quad \|P\bar{x} - Px - ey\| \leq qec(\|x\|)\|y\|$$

$$(S_2) \quad \|\bar{x} - x\| \leq \varepsilon B\|y\|$$

Main Result for Surjectivity

Theorem 9.38

Theorem 9.38: Let X and Y be Banach spaces. Let $P : \mathcal{D}(P) \subset X \rightarrow Y$ have closed graph. For $x \in \mathcal{D}(P)$, let $\Gamma(x)(P)$ denote the class (S). Suppose that y_0 is such that for each $x \in \mathcal{D}(P)$, the element $y_0 - Px$ belongs to the closure of the set $\Gamma(x)(P)$ defined by (S_1) and (S_2) . Then the equation $Px - y_0 = 0, x \in \mathcal{D}(P)$ has a solution.

Proof idea: Use the contraction property to show that an approximate solution sequence converges to an actual solution.

Application to Functional Equations

Caristi–Kirk Fixed Point Theorem

The generalized contractors framework allows us to apply:

Caristi–Kirk Fixed Point Theorem: Let X be a complete metric space. If $f : X \rightarrow X$ is a continuous map and there exists a continuous function $\varphi : X \rightarrow \mathbb{R}$ such that

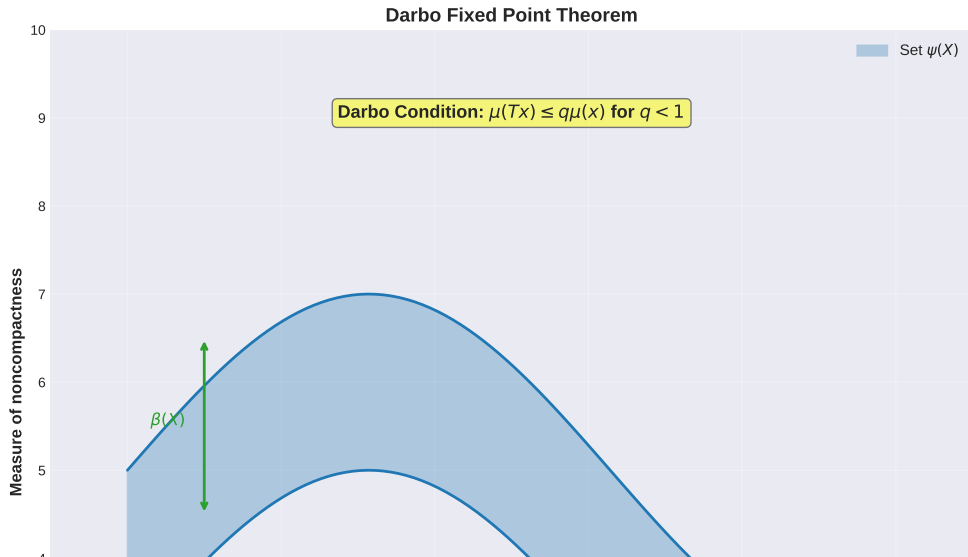
$$\rho(x, fx) \leq \varphi(x) - \varphi(fx) \quad \text{for all } x \in X$$

then f has a fixed point.

This generalizes Banach's contraction principle and applies to many nonlinear problems where classical contraction conditions fail.

Measure of Noncompactness

Darbo's Theorem Application



Summary of Key Fixed Point Theorems Used I

1 Banach Contraction Principle:

$$\|Tx - Ty\| \leq q\|x - y\|, \quad q < 1 \implies \exists! x^* : Tx^* = x^*$$

2 Darbo's Theorem (Measure of Noncompactness):

$$\mu(Tx) \leq q\mu(x), \quad q < 1 \implies \exists x^* : Tx^* = x^*$$

3 Caristi-Kirk Theorem:

$$\rho(x, fx) \leq \varphi(x) - \varphi(fx) \implies \exists x^* : fx^* = x^*$$

4 Class (S) Contractors (Surjectivity): Existence of solutions to nonlinear operator equations via generalized directional contractors.

Numerical Example

Volterra Integral Equation

Problem: Solve the Volterra integral equation

$$x(t) = t + \int_0^t (t-s)x(s)ds, \quad t \in [0, 1]$$

Python Implementation:

```
import numpy as np
from scipy.integrate import quad
import matplotlib.pyplot as plt

def volterra_kernel(t, s):
    return t - s

def fixed_point_iteration(t_vals, n_iter=20, tol=1e-6):
    x = np.zeros_like(t_vals)

    for iteration in range(n_iter):
        x_old = x.copy()
        x = t_vals.copy() # Initial term

        for i, t in enumerate(t_vals):
            integrand = lambda s: volterra_kernel(t, s) * np.interp(s, t_vals, x_old)
            integral, _ = quad(integrand, 0, t)
            x[i] += integral
```

Numerical Example Results

Error Behavior

Theoretical Result: By the Banach contraction principle:

- The operator T satisfies $\|Tx - Ty\| \leq q\|x - y\|$ with $q < 1$
- The error decreases exponentially: $\|e_n\| \leq q^n \|e_0\|$
- Solution converges to the unique fixed point x^*

Numerical observation: For $t \in [0, 1]$, the contraction constant $q \approx 0.5$, giving:

$$e_{n+1} \approx 0.5 \cdot e_n$$

This ensures rapid convergence in practice.

Practical Applications

These theoretical results have important applications:

- **Integral Equations:** Modeling nonlinear phenomena in physics and engineering
 - Population dynamics
 - Radiative transfer problems (Chandrasekhar equations)
 - Reaction-diffusion processes
- **Volterra Systems:** Integro-differential equations with memory effects
 - Viscoelasticity and material behavior
 - Biological systems with hereditary effects
 - Control systems with delay
- **Functional Equations:** General nonlinear mappings
 - Implicit function theorems
 - Inverse problems
 - Operator theory applications

Comparison of Fixed Point Methods

Method	Condition	Uniqueness	Scope
Banach	$\ Tx - Ty\ \leq q\ x - y\ $	Yes	Metric spaces
Darbo	$\mu(Tx) \leq q\mu(x)$	No	Measures of NCP
Caristi-Kirk	Energy decrease	Yes	Any complete space
Class (S)	Directional condition	Existence	Surjectivity

Each method has different requirements and provides different guarantees:

- **Uniqueness** is guaranteed by Banach but not by Darbo
- **Applicability** varies: some work on general metric spaces, others need norms
- **Computational** methods follow from Banach/Caristi-Kirk via iteration

Conclusions and Extensions I

Key Takeaways:

- 1 Fixed point theorems provide powerful tools for proving existence and uniqueness of solutions to integral and integro-differential equations
- 2 The Darbo condition allows application beyond classical contraction settings by using measures of noncompactness
- 3 Volterra systems model many real-world phenomena with memory and delay effects
- 4 Generalized directional contractors extend surjectivity results to broad classes of nonlinear operators
- 5 Numerical methods based on these theorems are implementable and convergent

Further Topics:

- Weak solutions and regularity theory
- Stability analysis of solutions
- Perturbation methods for integral equations
- Hybrid fixed point methods combining multiple approaches

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Thank You

Questions and Discussion

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